

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester V Examination, 2016-17

Botany**Paper: BOB-503 — Plant Ecology**

Time: Three Hours

Full Marks: 70

Note: Attempt five questions in all, selecting at least two questions from Section-A and Section-B. Question No. 1 is compulsory. The figures in the right-hand margin indicate marks.

Question 1. Answer the following in not more than 50 words each: [1 × 10 = 10]

- (i) Define ecosystem.
- (ii) Name two major greenhouse gases.
- (iii) Stratification in water body.
- (iv) What is soil profile?
- (v) Define carrying capacity.
- (vi) Decibel is a unit of what?
- (vii) Minamata disease is caused by _____.
- (viii) Photochemical smog.
- (ix) Give two examples of primary succession.
- (x) Humus.

SECTION – A

Question 2. What is a plant community? Discuss synthetic characters of a community. [15]

Question 3. Define population. Discuss various characters of population. [15]

Question 4. Describe the nitrogen cycle in detail. [15]

Question 5. Write critical notes on any two of the following: [7.5 × 2 = 15]

- (a) Light as ecological factor
- (b) Carbon cycle
- (c) Universal model of energy flow

SECTION – B

Question 6. Write critical comments on any two of the following: [7.5 × 2 = 15]

- (a) Eutrophication
- (b) Importance Value Index (IVI)
- (c) Concept of ecological niche

Question 7. Define ecological community. Discuss various theories of mechanism of ecological succession. [15]

Question 8. Write descriptive notes on any two of the following: [7.5 × 2 = 15]

- (a) Acid rain
- (b) Thermal pollution
- (c) Teratogens

Question 9. Define biodiversity. Discuss in brief indices of species diversity. [15]

[Click Here](#)

[Click Here](#)

[Click Here](#)